1 Data Structure and Identified Issues

The analysis is based on three datasets: `signals_raw_plus`, `msf`, and `factors_monthly`. The stock-level datasets (`signals_raw_plus` and `msf`) together form a monthly panel dataset indexed by PERMNO and time.

Number of Observations and Identifiers. The `signals_raw_plus` dataset contains 10,072 observations across 100 distinct PERMNOs, while the `msf` dataset contains 10,789 observations for the same set of securities. The `factors_monthly` dataset consists of 1,182 monthly observations and does not include PERMNO, as it represents aggregate factor returns.

Date Ranges. The `signals_raw_plus` and `msf` datasets both span from 1995-01-31 to 2024-12-31, with the frequency of a month, ensuring consistent coverage for return prediction. The `factors_monthly` dataset covers a longer period from 1926-07-01 to 2024-12-01, but only the overlapping sample is used in the analysis.

Dataset Contents and Required Fields. The `signals_raw_plus` dataset contains firm-level predictive signals indexed by **PERMNO** and **fdate**, including 126 numerical features used as candidate predictors. The `msf` dataset provides realized monthly returns (`RET`) along with identifiers **PERMNO** and **DATE**. The `factors_monthly` dataset includes Fama-French-Carhart factor returns (`mktrf`, `smb`, `hml`, `umd`) and the risk-free rate (`rf`). Together, these datasets contain all essential variables required for a return-prediction framework.

Duplicate Observations. No duplicate PERMNO–month observations are found in either the `signals_raw_plus` or `msf` datasets. Each stock-month pair is uniquely defined, ensuring that the panel structure is well-formed and suitable for merging.

Identifiers and Merge Keys. PERMNO is used as the primary permanent identifier due to its stability over time and consistency across CRSP-style datasets. Calendar month identifiers derived from `fdate` and `DATE` are used for temporal alignment. For the `factors_monthly` dataset, merging is performed on the month dimension only, as it does not contain stock-level identifiers. Variables such as prices, returns, or trading volume are not used as merge keys, as they do not uniquely identify observations.

Panel Structure and Timing. The stock-level panel is **unbalanced** because not all stocks are observed in every time period. Although there are 100 distinct PERMNOs, the total number of observations is smaller than the full PERMNO $\times$ time grid, reflecting entry (e.g., IPO) and exit (e.g., delisting) of firms. In contrast, the `factors_monthly` dataset is a pure time series and does not form a panel.

For a proper return-prediction framework, signals formed at time t must be used to predict returns at time $t + 1$, rather than contemporaneous returns. This timing structure ensures that only information available at time t is used, thereby avoiding look-ahead bias and maintaining economic validity.

2 Signal Selection and Economic Interpretation

2.1 Selected Signals

Signal	Category	Economic Idea	Prediction	Sign Adj.
Momentum	Momentum	Return continuation due to underreaction	$\uparrow \rightarrow \uparrow$	No
BtM	Value	Cheap stocks earn value premium	$\uparrow \rightarrow \uparrow$	No

Table 1: Momentum and Value signals.

Signal	Category	Economic Idea	Prediction	Sign Adj.
ROA	Profitability	More profitable firms perform better	$\uparrow \rightarrow \uparrow$	No
ivol	Risk	High idiosyncratic risk linked to mispricing	$\uparrow \rightarrow \downarrow$	Yes ($-$`ivol`)

Table 2: Profitability and Risk signals.

3 Return-Prediction Results

Signal	Mean LS	Ann. Vol	Sharpe	CAPM α	FF4 α	SE	Max DD	Avg N (L/S)
Momentum	-0.0003	0.6621	-0.01	0.0058	-0.0043	0.0096	-1.01	5.90 / 5.05
BtM	0.0229	0.5940	0.46	0.0229	0.0231	0.0091	-0.85	5.64 / 4.82
ROA	0.0004	0.6509	0.01	0.0040	0.0010	0.0100	-1.08	5.85 / 4.96
ivol	0.0184	0.5995	0.37	0.0087	0.0141	0.0085	-0.93	5.87 / 5.06
Composite	0.0158	0.7296	0.26	0.0184	0.0148	0.0114	-1.08	5.47 / 4.78

Table 3: Performance of single-signal and composite long–short portfolios.

Summary The results show that return predictability varies significantly across signals. The BtM signal delivers the strongest performance, with the highest average long–short return, Sharpe ratio, and statistically meaningful four-factor alpha. The ivol signal also generates positive returns, although with weaker statistical significance. In contrast, momentum and ROA exhibit negligible predictive power. However, these results should be interpreted with caution, as the sample includes only 100 stocks and exhibits high volatility and severe drawdowns, limiting diversification and increasing noise. The composite signal produces stable but weaker performance relative to BtM, with lower Sharpe ratio and higher volatility. This suggests that simple averaging of signals does not improve performance, as weaker predictors dilute stronger ones in a small-sample setting.

4 Interpretation of Results

The results should be interpreted with caution due to several limitations in the sample design. First, the analysis is based on a small cross-section of only 100 stocks, which reduces statistical power and limits diversification. Second, the absence of industry controls may lead signals to capture sector effects rather than true cross-sectional predictability. Third, the use of equal-weighted portfolios overemphasizes small-cap stocks and may overstate returns relative to realistic implementations.

In addition, the composite signal does not improve performance, as weaker predictors such as momentum and ROA dilute the contribution of stronger signals like BtM. The backtest also ignores transaction costs and turnover, which could significantly reduce profitability. Overall, while the results provide useful insights into signal behavior, they are not sufficiently robust for reliable investment conclusions.

5 Next Steps with the Full CRSP/Compustat Universe

To extend this analysis to the full CRSP/Compustat universe, several improvements in data coverage and methodology are required. First, the cross-sectional sample should be expanded from 100 stocks to the full investable universe, which would significantly improve statistical power and diversification. Second, firm-level fundamentals from Compustat should be carefully merged with CRSP data using a proper linking table (e.g., CCM link), ensuring accurate alignment of accounting and market data.

In addition, stricter data filters should be applied, such as excluding micro-cap stocks (as they are quite uncertain and high-risked), requiring minimum price thresholds, and handling delisting returns explicitly. Industry classification should also be incorporated to control for sector effects and enable industry-neutral portfolio construction. Finally, more robust preprocessing steps, including winsorization and consistent handling of missing values, should be implemented to improve signal stability.

6 Improvements for an Institutional-Grade Backtesting System

To build an institutional-grade backtesting system, several enhancements are necessary:

- **Transaction costs and market frictions:** Explicitly model transaction costs, bid–ask spreads, and turnover, as these can materially reduce strategy profitability.
- **Portfolio construction:** Move beyond equal-weighting to value-weighted or risk-controlled approaches, better reflecting real capital allocation.
- **Signal construction:** Improve signals through cross-sectional standardization and weighting based on predictive strength (e.g., IC or Sharpe), and consider more advanced combination methods beyond simple averaging.
- **Robustness and validation:** Perform out-of-sample testing and robustness checks (e.g., subperiod analysis) to avoid overfitting and data-snooping bias.

These enhancements would make the backtesting framework more realistic, stable, and suitable for practical deployment.